CS 6012: Autumn 2026 — Open Project

Intermediate Checkpoint #4: Status Report (Submission 2: Code-Based Project)

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1. Title of the Project

GNN Molecular Substructure Search: Accuracy vs. Speed Trade-offs Against Classical Fingerprint Screening

2. Updates on Problem Statement & Cascaded Screening Architecture

Exact subgraph isomorphism matching ($Q \subseteq T$) via combinatorial algorithms (VF2/Ullmann in RDKit) scales exponentially in the worst case $O(|V_T|^{|V_Q|})$ and becomes computationally prohibitive across large chemical discovery libraries (50,000+ molecules). Because pure neural classifiers suffer from false negatives that can discard critical bioactive scaffolds, we have formally refined our pipeline into a **two-stage cascaded filtering paradigm**: (1) A bond-aware Graph Isomorphism Network with NeuroMatch-style order embeddings ($z_Q \leq z_T$ element-wise in $\blacksquare_{\geq 0}^{64}$) rapidly screens the database in <3 ms on GPU to shortlist candidates at a strict high-recall constraint (≥98%), and (2) RDKit's exact HasSubstructMatch is evaluated *only* on the shortlisted pool, guaranteeing 100% precision with zero false positives. The objective is quantifying the Pareto frontier of this neural cascade against the industrial gold standard: RDKit's 2048-bit Chem.PatternFingerprint bitwise screen followed by exact verification.

3. Empirical Results & Difficulties Encountered

Retrieval Method	Precision	Recall	F1-Score	Latency / Query	Speedup	Avg. Candidates
RDKit Exact (Brute Force)	1.000	1.000	1.000	31.84 ms	1.0×	494.2
Pattern FP + RDKit (Baseline)	1.000	1.000	1.000	**5.61 ms**	**5.7×**	703.3
GNN B (Order Embedding alone)	0.270	0.771	0.399	**2.79 ms**	11.4×	1,413.9
Cascade (GNN B Shortlist + RDKit)	1.000	0.982	0.991	20.21 ms	1.6×	485.5

- **Shortlist Overhead & Verification Bottleneck:** GNN B is over 2× faster at vector screening than the fingerprint screen (2.79 ms vs. 1.09 ms). However, at the ≥98% recall threshold, its false positive rate produces ~1,414 candidates (vs. 703 for Pattern FP). At ~0.05 ms per RDKit check, verifying false positives creates an execution bottleneck, lowering cascaded speedup from a theoretical 11× to 1.6×.
- **Query-Size Receptive Field Decay (4 vs. 12 Atoms):** Empirical breakdown shows strong recall on small fragments (4 atoms: 83.4%; 5 atoms: 83.7%), but sharp degradation on larger subgraphs (9 atoms: 67.8%; 10 atoms: 28.4%; 12 atoms: 43.6%). The 4-layer GINEConv receptive field and global sum-pooling struggle to distinguish long-range cycle topologies as molecular diameter expands.
- **CPU Bottleneck in Hard-Negative Mining:** Constructing balanced training pairs requiring hard negatives (pairs passing bitwise fingerprint screens but failing exact topological matching) requires >40,000 CPU-bound RDKit exact calls during batching.

4. Contributions of Members

- **Aayush Gupta (25m0140):** GPU acceleration and CUDA runtime integration on RTX 3050, PyG GINEConv message-passing optimization, order-embedding max-margin loss violation tuning, threshold calibration curves for the cascade shortlist, and Checkpoint documentation.
- **Srijal Gupta (25m0387):** ZINC250k query generation stratified across atom sizes (4–12 atoms + 20 functional groups), hard-negative sampling pipeline, classical Pattern Fingerprint benchmarking suite, error analysis by query size, and schedule tracking.
- **Joint Deliverables:** Cascaded pipeline evaluation, error analysis documentation, presentation slide preparation, and repository maintenance.

5. Codebase Packaging & Repository Submission Progress

- **Codebase & Artifact Readiness:** The complete standalone Python pipeline (gnn_substructure_search.py), interactive Colab notebook (GNN_Substructure_Search.ipynb), benchmark results suite (results/pilot/), tradeoff plots, and model configurations are fully developed and tested locally.
- **Repository Submission Status:** All core files, scripts, and documentation have been structured and verified. Final repository integration and upload to the designated course repository are actively in progress and will be delivered alongside the final submission package.
- **Updated Schedule Spreadsheet:** Submitted as companion PDF (Updated_Schedule_Spreadsheet.pdf) and online at: Google Spreadsheet Link.